MATH INQUIRERS HOLIDAY PROGRAM

Weekly Email Pack

Parent and tutor emails for the five weeks of the Level 6 (ages 11-13) holiday program

Notice it. Test it. Explain why.

How to use this pack

- Send the parent email and the Week N Student Pages PDF the day before the first lesson of that week.
- Send the tutor email with the Week N Tutor Scripts and Answers PDF at least two days before. Tutors only; the answers must not go to parents.
- Put parents in BCC so no one sees another family's email address. Get parents' consent to email before you start (PDPA).
- Replace the words in [square brackets] before sending.
- The PDFs are in the weekly folder (Math_Inquirers_Level6_Weekly_PDFs.zip): Week1_Student_Pages.pdf to Week5_Student_Pages.pdf and Week1_Tutor_Scripts_and_Answers.pdf to Week5_Tutor_Scripts_and_Answers.pdf. The Week 1 student PDF also holds the inquiry journal page and all the cut-outs; the Week 5 PDF ends with the certificate.
- The slide deck links in the tutor emails are private to you. Share each deck with the tutor first, or send them the file.
- This level is online-first and children code on a computer, so the Week 1 parent email also asks for a laptop or desktop with Scratch open. Attach the Scratch warm-up handout to the Week 1 email if you have it ready.

Week 1: Lessons 1 and 2

Parent email

Subject: Math Inquirers Level 6 (ages 11-13) Week 1: Sequences and the nth Term and Primes, Factors and Prime Factorisation

Attachment: Week1_Student_Pages.pdf

Dear parents,

This week our Math Inquirers Level 6 (ages 11-13, after the PSLE) meet two topics, in lessons 1 and 2.

Lesson 1, Code the Pattern (Sequences and the nth Term). Finding the rule of a number pattern, testing it in a short program, and finding the 100th term without listing them all.

Lesson 2, Prime Hunters (Primes, Factors and Prime Factorisation). Prime numbers, factors, factor trees, and a program that counts factors.

Your child's workbook pages for this week are attached. Please print them single-sided if you can. Children will work on the pages in class, so there is no need to finish them in advance.

The cut-out pages are at the end of the Week 1 PDF (Unplugged code cards, square grid). Please cut them out and keep them in one envelope; we use them through the program. The inquiry journal page is at the front. This week's lessons use: unplugged code cards. Each child needs a laptop or desktop with internet for the coding parts (a tablet works for Scratch but is slower). Scratch is free at scratch.mit.edu and no account is needed; please let your child open it once before the first lesson. [Attach the Scratch warm-up handout if you have it.]

One thing to try at home (5 minutes): Make a matchstick pattern of triangles or squares. Ask how many sticks the 10th figure needs before you build it.

Ask your child: "Can you test your rule on the next figure?" or "What would you try next if it is not divisible by 2, 3 or 5?"

We are not looking for speed. We are looking for a child who can say what they noticed, test it, and explain why. If your child is stuck, please ask "What do you notice?" before you give the answer.

Warm regards,
[Your name]
Preciouslyours Pte Ltd
[Contact number]

Week 1: Tutor email

Subject: Math Inquirers Level 6 (ages 11-13) Week 1 tutor pack: Lessons 1 and 2

Attachments: Week1_Tutor_Scripts_and_Answers.pdf, Week1_Student_Pages.pdf

Hi [tutor name],

Here is your pack for week 1: Code the Pattern; Prime Hunters.

Attached are the minute-by-minute scripts with answers for both lessons, and the student pages. The slide decks are linked below.

- Lesson 1 deck: https://claude.ai/artifact/AL5fVRN59kqmMdhNzCkVGo
- Lesson 2 deck: https://claude.ai/artifact/QsahJ4TqnX9utsUQ83u1fC

Before the lesson, please:

1. Read both scripts once and try the Go further questions yourself.
2. Check your materials are ready. Lesson 1: Laptop or desktop with Scratch 3 open (scratch.mit.edu, no account needed), about 25 matchsticks or toothpicks per child (or squared paper), mini whiteboard, workbook. Optional: a Python editor for children who already know Python. Lesson 2: Laptop or desktop with Scratch 3 open, a printed 1 to 100 grid (or squared paper), two coloured pencils, mini whiteboard, workbook. Optional: a Python editor for children who already know Python.
3. After each lesson, fill in the Progress Notes for each child (30 seconds each) and send me the sheet.

Please do not forward the answer pages to parents.

Thank you,
[Your name]

Week 2: Lessons 3 and 4

Parent email

Subject: Math Inquirers Level 6 (ages 11-13) Week 2: Negative Numbers and Expressions and Equations

Attachment: Week2_Student_Pages.pdf

Dear parents,

This week our Math Inquirers Level 6 (ages 11-13, after the PSLE) meet two topics, in lessons 3 and 4.

Lesson 3, Below Zero (Negative Numbers). Negative numbers on a number line, temperatures and floors below zero, and programs that go up and down.

Lesson 4, Algebra Machines (Expressions and Equations). Number machines and simple algebra: finding the rule, working backwards, and a program that repeats until it hits a target.

Your child's workbook pages for this week are attached. Please print them single-sided if you can. Children will work on the pages in class, so there is no need to finish them in advance.

Please keep the Week 1 envelope handy. This week's lessons use: unplugged code cards.

One thing to try at home (5 minutes): Look at the weather in a cold city and ask for the difference between its low and ours.

Ask your child: "Which way do you move on the number line?" or "What is the opposite of each step?"

We are not looking for speed. We are looking for a child who can say what they noticed, test it, and explain why. If your child is stuck, please ask "What do you notice?" before you give the answer.

Warm regards,
[Your name]
Preciouslyours Pte Ltd
[Contact number]

Week 2: Tutor email

Subject: Math Inquirers Level 6 (ages 11-13) Week 2 tutor pack: Lessons 3 and 4

Attachments: Week2_Tutor_Scripts_and_Answers.pdf, Week2_Student_Pages.pdf

Hi [tutor name],

Here is your pack for week 2: Below Zero; Algebra Machines.

Attached are the minute-by-minute scripts with answers for both lessons, and the student pages. The slide decks are linked below.

- Lesson 3 deck: https://claude.ai/artifact/XZD6Utf7JY7EsWsTPK4KgB
- Lesson 4 deck: https://claude.ai/artifact/FvMdour8GjREuxrJ8HQVCs

Before the lesson, please:

1. Read both scripts once and try the Go further questions yourself.
2. Check your materials are ready. Lesson 3: Laptop or desktop with Scratch 3 (scratch.mit.edu, free, no log-in needed), mini whiteboard and marker, workbook, pencil. Lesson 4: Laptop or desktop with Scratch 3 (scratch.mit.edu, free), mini whiteboard and marker, workbook, pencil.
3. After each lesson, fill in the Progress Notes for each child (30 seconds each) and send me the sheet.

Please do not forward the answer pages to parents.

Thank you,
[Your name]

Week 3: Lessons 5 and 6

Parent email

Subject: Math Inquirers Level 6 (ages 11-13) Week 3: Counting and Probability

Attachment: Week3_Student_Pages.pdf

Dear parents,

This week our Math Inquirers Level 6 (ages 11-13, after the PSLE) meet two topics, in lessons 5 and 6.

Lesson 5, Every Possible Way (Counting). Counting every possible way: tree diagrams, the multiplication rule, handshakes, and nested loops.

Lesson 6, Fair or Unfair? (Probability). Probability as a fraction, fair and unfair games, and a program that rolls a die 1000 times.

Your child's workbook pages for this week are attached. Please print them single-sided if you can. Children will work on the pages in class, so there is no need to finish them in advance.

No cut-outs are needed this week. Please keep the Week 1 envelope handy.

One thing to try at home (5 minutes): Ask how many outfits can be made from 3 tops and 4 skirts, then draw all of them.

Ask your child: "How do you know you have not missed or counted twice?" or "Why is it close, but not exactly what we expected?"

We are not looking for speed. We are looking for a child who can say what they noticed, test it, and explain why. If your child is stuck, please ask "What do you notice?" before you give the answer.

Warm regards,
[Your name]
Preciouslyours Pte Ltd
[Contact number]

Week 3: Tutor email

Subject: Math Inquirers Level 6 (ages 11-13) Week 3 tutor pack: Lessons 5 and 6

Attachments: Week3_Tutor_Scripts_and_Answers.pdf, Week3_Student_Pages.pdf

Hi [tutor name],

Here is your pack for week 3: Every Possible Way; Fair or Unfair?.

Attached are the minute-by-minute scripts with answers for both lessons, and the student pages. The slide decks are linked below.

- Lesson 5 deck: https://claude.ai/artifact/H1azj3qWyF12pYYKYUKJSs
- Lesson 6 deck: https://claude.ai/artifact/2NecdycCDKH89oxzh8hwgE

Before the lesson, please:

1. Read both scripts once and try the Go further questions yourself.
2. Check your materials are ready. Lesson 5: Laptop or desktop with Scratch 3 (scratch.mit.edu) open, the starter project “Every Order” (one loop that says each tea), mini whiteboards, digit cards 0 to 9, squared paper. Lesson 6: Laptop or desktop with Scratch 3 (scratch.mit.edu), two dice per pair (or the tutor’s dice program), a coin, coloured counters and a paper bag, mini whiteboards, tally sheet.
3. After each lesson, fill in the Progress Notes for each child (30 seconds each) and send me the sheet.

Please do not forward the answer pages to parents.

Thank you,
[Your name]

Week 4: Lessons 7 and 8

Parent email

Subject: Math Inquirers Level 6 (ages 11-13) Week 4: Angles in Polygons and Clock (Modular) Arithmetic

Attachment: Week4_Student_Pages.pdf

Dear parents,

This week our Math Inquirers Level 6 (ages 11-13, after the PSLE) meet two topics, in lessons 7 and 8.

Lesson 7, Turtle Geometry (Angles in Polygons). Turtle geometry: turning angles, regular polygons and stars, and total turning.

Lesson 8, Secret Codes (Clock (Modular) Arithmetic). Secret codes: clock arithmetic, shift codes, cracking a code, and a program that scrambles a message.

Your child's workbook pages for this week are attached. Please print them single-sided if you can. Children will work on the pages in class, so there is no need to finish them in advance.

Please keep the Week 1 envelope handy. This week's lessons use: unplugged code cards, square grid.

One thing to try at home (5 minutes): Walk a regular shape on the floor: five steps, turn, repeat. How much did you turn in total when you came back?

Ask your child: "Why does the turtle always turn 360 degrees in total?" or "What happens when you go past Z?"

We are not looking for speed. We are looking for a child who can say what they noticed, test it, and explain why. If your child is stuck, please ask "What do you notice?" before you give the answer.

Warm regards,
[Your name]
Preciouslyours Pte Ltd
[Contact number]

Week 4: Tutor email

Subject: Math Inquirers Level 6 (ages 11-13) Week 4 tutor pack: Lessons 7 and 8

Attachments: Week4_Tutor_Scripts_and_Answers.pdf, Week4_Student_Pages.pdf

Hi [tutor name],

Here is your pack for week 4: Turtle Geometry; Secret Codes.

Attached are the minute-by-minute scripts with answers for both lessons, and the student pages. The slide decks are linked below.

- Lesson 7 deck: https://claude.ai/artifact/Wcr5DmgwQ4FZaHkEGchzop
- Lesson 8 deck: https://claude.ai/artifact/TngoHLTWb4WgFGegEdQKaw

Before the lesson, please:

1. Read both scripts once and try the Go further questions yourself.
2. Check your materials are ready. Lesson 7: A laptop or desktop with Scratch 3 open (scratch.mit.edu) and the Pen extension added, mini whiteboards, a protractor, plain or squared paper. Lesson 8: A laptop or desktop with Scratch 3 open (the tutor builds and shares a starter project with a list called alphabet holding A to Z; no .sb3 file exists in this repository, so build it beforehand), mini whiteboards, paper for code strips, scissors.
3. After each lesson, fill in the Progress Notes for each child (30 seconds each) and send me the sheet.

Please do not forward the answer pages to parents.

Thank you,
[Your name]

Week 5: Lessons 9 and 10

Parent email

Subject: Math Inquirers Level 6 (ages 11-13) Week 5: Averages and Data and Mixed: Maths and Code

Attachment: Week5_Student_Pages.pdf

Dear parents,

This week our Math Inquirers Level 6 (ages 11-13, after the PSLE) meet two topics, in lessons 9 and 10.

Lesson 9, Data Detectives (Averages and Data). Data: mean, median and mode, what changes the mean, and a program that finds the mean of a list.

Lesson 10, Showcase Challenge (Mixed: Maths and Code). Showcase: a mixed challenge round, then planning, building and testing a small puzzle program.

Your child's workbook pages for this week are attached. Please print them single-sided if you can. Children will work on the pages in class, so there is no need to finish them in advance.

Please keep the Week 1 envelope handy. This week's lessons use: square grid. The Certificate of Completion is the last page of the Week 5 PDF.

One thing to try at home (5 minutes): Collect the heights or pocket money of your family and work out the mean and the median. Which describes them better?

Ask your child: "When is the mean misleading?" or "How did you test it? What would you improve?"

We are not looking for speed. We are looking for a child who can say what they noticed, test it, and explain why. If your child is stuck, please ask "What do you notice?" before you give the answer.

Warm regards,
[Your name]
Preciouslyours Pte Ltd
[Contact number]

Week 5: Tutor email

Subject: Math Inquirers Level 6 (ages 11-13) Week 5 tutor pack: Lessons 9 and 10

Attachments: Week5_Tutor_Scripts_and_Answers.pdf, Week5_Student_Pages.pdf

Hi [tutor name],

Here is your pack for week 5: Data Detectives; Showcase Challenge.

Attached are the minute-by-minute scripts with answers for both lessons, and the student pages. The slide decks are linked below.

- Lesson 9 deck: https://claude.ai/artifact/Mnqm54v82yxtGBnqFviR9b
- Lesson 10 deck: https://claude.ai/artifact/PNoQzJNbyP4EBZpVHYAgSc

Before the lesson, please:

1. Read both scripts once and try the Go further questions yourself.
2. Check your materials are ready. Lesson 9: Laptop or desktop with Scratch 3 (scratch.mit.edu; free, in the browser), workbook, pencil, mini whiteboard, about 25 linking cubes or counters for levelling towers. A calculator only for checking. Lesson 10: Laptop or desktop with Scratch 3 (scratch.mit.edu), workbook (challenge round on page 1, planning sheet on page 2), pencil, mini whiteboard, the starter projects as .sb3 files (A Mystery Machine, B Prime or Not?, C Lucky Seven, D Clock Jumper; children use File, Load from your computer), or build them from the slides, certificates, stamps.
3. After each lesson, fill in the Progress Notes for each child (30 seconds each) and send me the sheet.

Please do not forward the answer pages to parents.

Thank you,
[Your name]

Week 5 closing email to parents

Subject: Math Inquirers Level 6 (ages 11-13): thank you, and your child's certificate

Dear parents,

Thank you for joining us for the Math Inquirers Holiday Program, Level 6 (ages 11-13, after the PSLE). Your child's Certificate of Completion is the last page of the Week 5 PDF. Please print it and add your child's name.

We would love your feedback in two minutes: [feedback form link]. What did your child enjoy most? What should we change?

If your child showed a love for puzzles, we may invite them to a short challenge session in the next holiday. We will contact you if so.

Warm regards,
[Your name]